Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON ROOTS & FORMATION OF QUADRATIC EQUATION

1. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as m/n then find the value of $(m + n)$.

माना द्विघात समीकरण $x^2 + 3x - k = 0$ के मूल a, b तथा $x^2 + 3x - 10 = 0$ के मूल c, d इस प्रकार है कि प्रथम समीकरण के मूलो के अन्तर का परिमाण द्वितीय समीकरण के मूलो के अन्तर के परिमाण का दुगुना है। यदि 'k' का मान m/n के रूप में व्यक्त किया जाए जहाँ m, n का HCF 1 हो तो $(m + n)$ मान क्या होगा?

Ans. 191

Sol. $x^2 + 3x - k = 0$

a, b are the roots

$$|a-b| = \left|\frac{\sqrt{D}}{a}\right| = \left|\frac{\sqrt{9+4k}}{1}\right| = \sqrt{9+4k}$$

$x^2 + 3x - 10 = 0$

c, d are the roots.

$$|c-d| = \left|\frac{\sqrt{9+40}}{1}\right| = \left|\sqrt{49}\right|$$

$\because \sqrt{9+4k} = 2\cdot\sqrt{49}$

$9 + 4k = 4 \times 49$

$4k = 196 - 9$

$4k = 187$

$$k = \frac{187}{4} = \frac{m}{n}$$

$\Rightarrow m + n = 187 + 4 = 191$

2. A quadratic polynomial $f(x) = x^2 + ax + b$ is formed with one of its zeros being $\dfrac{4+3\sqrt{3}}{2+\sqrt{3}}$ where a and b are integers. Also $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ is a biquadratic polynomial such that $g\left(\dfrac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3} + d$ where c and d are also integers. Find the values of a, b, c and d.

एक द्विघात बहुपद $f(x) = x^2 + ax + b$ इस प्रकार बनाया गया है कि इसका एक शून्य $\dfrac{4+3\sqrt{3}}{2+\sqrt{3}}$ है जहाँ a तथा b पूर्णांक है। साथ ही $g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$ इस प्रकार है कि $g\left(\dfrac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3} + d$ जहाँ c तथा d भी पूर्णांक है। a, b, c एवं d का मान निकालिए।

Ans. $a = 2, b = -11, c = 4, d = -1$

Sol. $f(x) = x^2 + ax + b$

$$\alpha = \frac{4+3\sqrt{3}}{2+\sqrt{3}} = \frac{\left(4+3\sqrt{3}\right)\left(2-\sqrt{3}\right)}{\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)} = \frac{8-4\sqrt{3}+6\sqrt{3}-9}{4-3} = 2\sqrt{3}-1$$

then β will be $\left(-2\sqrt{3}-1\right)$

sum of roots $= -a$

$2\sqrt{3}-1-2\sqrt{3}-1 = -a$

$a = 2$

Product of roots $= b$

$-\left(2\sqrt{3}-1\right)\left(2\sqrt{3}+1\right) = b$

$-(12-1) = b \Rightarrow b = -11$

$\because g\left(\dfrac{4+3\sqrt{3}}{2+\sqrt{3}}\right) = c\sqrt{3}+d$

$g\left(2\sqrt{3}-1\right) = c\sqrt{3}+d$, $\qquad g(x) = x^4 + 2x^3 - 10x^2 + 4x - 10$

$\because x = 2\sqrt{3}-1$

$x+1 = 2\sqrt{3}$

$x^2 + 1 + 2x = 12$

$x^2 + 2x - 11 = 0$

Now, dividing $x^4 + 2x^3 - 10x^2 + 4x - 10$ by $x^2 + 2x - 11$, we get $x^2 + 1$ as quotient and $2x + 1$ as remainder.

$\therefore g(x) = (x^2 + 2x + 11)(x^2 + 1) + (2x + 1)$

$g(x) = 0 + (2x + 1)$

$\because g\left(2\sqrt{3}-1\right) = c\sqrt{3}+d$

$2\cdot\left(2\sqrt{3}-1\right)+1 = c\sqrt{3}+d$

$4\sqrt{3}-2+1 = c\sqrt{3}+d$

$4\sqrt{3}-1 = c\sqrt{3}+d$

$c = 4, d = -1$

3. α, β are the roots of the equation $K(x^2-x)+x+5=0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta)+(\beta/\alpha)=4/5$. Find the value of $(K_1/K_2)+(K_2/K_1)$.

α, β समीकरण $K(x^2-x)+x+5=0$ के मूल है। यदि K_1 तथा K_2, K के दो मान है जिसके लिये α, β सम्बन्ध $(\alpha/\beta)+(\beta/\alpha)=4/5$ को सन्तुष्ट करते है तो $(K_1/K_2)+(K_2/K_1)$ का मान बताइयें।

Ans. 254

Sol. $k\left(x^2-x\right)+x+5=0\left\langle\begin{matrix}\alpha\\ \beta\end{matrix}\right.$

$kx^2-kx+x+5=0$

$kx^2-x(k-1)+5=0$

$\alpha+\beta=\dfrac{k-1}{k};\alpha\cdot\beta=\dfrac{5}{k}$

$\because \dfrac{\alpha}{\beta}+\dfrac{\beta}{\alpha}=\dfrac{4}{5}$

$\dfrac{\alpha^2+\beta^2}{\alpha\beta}=\dfrac{4}{5}$

$5\cdot\left\{(\alpha+\beta)^2-2\alpha\beta\right\}=4\alpha\beta$

$5\cdot\left\{\left(\dfrac{k-1}{k}\right)^2-\dfrac{10}{k}\right\}=4\cdot\dfrac{5}{k}$

$k\neq 0$

$k^2+1-2k-10k=4k$

$k^2-16k+1=0\left\langle\begin{matrix}k_1\\ k_2\end{matrix}\right.$

$k_1+k_2=16$

$k_1\cdot k_2=1$

Now $\dfrac{k_1}{k_2}+\dfrac{k_2}{k_1}=\dfrac{k_1^2+k_2^2}{k_1\cdot k_2}=\dfrac{(k_1+k_2)^2-2k_1\cdot k_2}{k_1\cdot k_2}$

$=\dfrac{16^2-2\times 1}{1}=254$

CONDITION OF COMMON ROOTS

4. If the quadratic equations, $x^2+bx+c=0$ and $bx^2+cx+1=0$ have a common root then prove that either $b+c+1=0$ or $b^2+c^2+1=bc+b+c$.

यदि द्विघात समीकरणों $x^2+bx+c=0$ तथा $bx^2+cx+1=0$ का एक उभयनिष्ठ मूल हो तो सिद्ध कीजिये या तो $b+c+1=0$ या $b^2+c^2+1=bc+b+c$ होगा।

Ans.

Sol. $x^2 + bx + c = 0$

α, β are the roots.

$bx^2 + cx + 1 = 0$

α, γ are the roots.

$\alpha^2 + b\alpha + c = 0$

$b\alpha^2 + c\alpha + 1 = 0$

$$\frac{\alpha^2}{b-c^2} = \frac{\alpha}{bc-1} = \frac{1}{c-b^2}$$

when $\dfrac{\alpha}{bc-1} = \dfrac{1}{c-b^2}$

$$\alpha = \frac{bc-1}{c-b^2}$$

and when $\dfrac{\alpha^2}{b-c^2} = \dfrac{\alpha}{bc-1}$

$$\alpha = \frac{(b-c^2)}{(bc-1)}$$

$$\Rightarrow \frac{(bc-1)}{c-b^2} = \frac{(b-c^2)}{(bc-1)}$$

$(bc-1)^2 = (b-c^2)(c-b^2)$

$b^2c^2 + 1 - 2bc = bc - b^3 - c^3 + b^2c^2$

$b^3 + c^3 + 1^3 = 3bc \times 1$

$\therefore (b + c + 1)\cdot(b^2 + c^2 + 1 - bc - c - b) = 0$

$\Rightarrow b + c + 1 = 0$ OR $b^2 + c^2 + 1 = bc + c + b$

Hence proved.

5. Find the value of m for which the quadratic equations $x^2 - 11x + m = 0$ & $x^2 - 14x + 2m = 0$ may have common root.

'm' का वह मान बताइये जिसके लिये द्विघात समीकरणों $x^2 - 11x + m = 0$ तथा $x^2 - 14x + 2m = 0$ के उभयनिष्ठ मूल हों।

Ans. 0, 24

Sol. $x^2 - 11x + m = 0$ ...(i)

$x^2 - 14x + 2m = 0$ ...(ii)

from (i) – (ii)

$3x - m = 0$

$$x = \frac{m}{3}$$

Now $\frac{m^2}{9} - \frac{11m}{3} + m = 0$
$m^2 - 33m + 9m = 0$
$m^2 - 24m = 0$
$m = 0, 24$

6. Find the product of uncommon real roots of the two polynomials $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and $Q(x) = x^3 + 4x^2 - x - 10$.

बहुपदों $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ तथा $Q(x) = x^3 + 4x^2 - x - 10$ के उन मूलो का गुणनफल ज्ञात कीजिए जो उभयनिष्ठ नही है।

Ans. 6

Sol. $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$
$Q(x) = x^3 + 4x^2 - x - 10$
$= x^3 + 2x^2 + 2x^2 + 4x - 5x - 10$
$= x^2 (x + 2) + 2x (x + 2) - 5 (x + 2)$
$= (x + 2) (x^2 + 2x - 5)$
$P(x) \div x^2 + 2x - 5$
We get $x^2 - 3$ as quotient and 0 as remiander,.
$\Rightarrow Q(x) = (x^2 - 3) (x^2 + 2x - 5)$
$= \left(x - \sqrt{3}\right)\left(x + \sqrt{3}\right)\left(x^2 + 2x - 5\right)$
$\therefore$ Uncommon roots $= -2,\ \sqrt{3},\ -\sqrt{3}$
Product $= -2 \times \sqrt{3} \times \left(-\sqrt{3}\right)$
$= 6$

RANGE OF RATIONAL FUNCTIONS

7. Let α, β be real roots of the quadratic equation $x^2 - kx + k^2 + k - 5 = 0$. If m and M are respectively the minimum and maximum value of $\alpha^2 + \beta^2$, then find $(m + M)$.

माना द्विघात समीकरण $x^2 - kx + k^2 + k - 5 = 0$ के वास्तविक मूल α, β है। यदि $\alpha^2 + \beta^2$, के न्यूनतम एवं उच्चतम मान क्रमशः m तथा M है तो $(m + M)$ का मान ज्ञात कीजिये।

Ans. 13

Sol. $x^2 - kx + k^2 + k - 5 = 0$
α, β are the roots.
$\alpha + \beta = k$
$\alpha\beta = (k^2 + k - 5)$
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
$= k^2 - 2 (k^2 + k - 5)$
$= k^2 - 2k^2 - 2k + 10$

$= -k^2 - 2k + 10$

$\because$ roots are real of Quadratic Equation

$\therefore D \geq 0$

$k^2 - 4(k^2 + k - 5) \geq 0$

$k^2 - 4k^2 - 4k + 20 \geq 0$

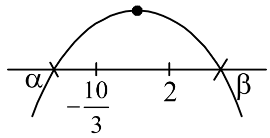
$-3k^2 - 4k + 20 \geq 0$

$3k^2 + 4k - 20 \leq 0$

$3k^2 + 10k - 6k - 20 \leq 0$

$k(3k + 10) - 2(3k + 10) \leq 0$

$(3k + 10)(k - 2) \leq 0$



$$-\frac{10}{3} \leq k \leq 2$$

from

$\alpha^2 + \beta^2 = -k^2 - 2k + 10 = f(k)$

$a < 0,$

$D = 4 + 40 = 44 > 0$

$$k = \frac{-2 + \sqrt{44}}{2} = 2\cdot35 \text{ or } -4\cdot35$$

$$\frac{-b}{2a} = \frac{2}{-2} = -1$$

$$\frac{-D}{4a} = \frac{-44}{4 \times (-1)} = 11$$

$M = 11$

$$f\left(\frac{-10}{3}\right) = \frac{-100}{9} + \frac{20}{3} + 10 = \frac{-100 + 60 + 90}{9} = +\frac{50}{9}$$

$f(2) = -4 - 4 + 10 = 2 = m$

$\therefore m + M = 2 + 11 = 13$

ALWAYS POSITIVE & ALWAYS NEGATIVE BASED QUESTIONS

8. Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a + b)x + (a - b + 8) = 0$ has unequal real roots for all $a \in R$.

माना a, b कोई वास्तविक संख्याऐं है। वह सबसे छोटी प्राकृतिक संख्या 'b' ज्ञात कीजिए जिसके लिये समीकरण $x^2 + 2(a + b)x + (a - b + 8) = 0$ के, सभी $a \in R$ के लिये, असमान वास्तविक मूल हो।

Ans. 5

Sol. $x^2 + 2(a + b)x + (a - b + 8) = 0$

$D > 0$

$4(a + b)^2 - 4(a - b + 8) > 0$

$a^2 + b^2 + 2ab - a + b - 8 > 0$

$a^2 + (2b - 1)a + (b^2 + b - 8) > 0$

$\Rightarrow D < 0$

$(2b-1)^2 - 4(b^2 + b - 8) < 0$
$4b^2 + 1 - 4b - 4b^2 - 4b + 32 < 0$
$-8b + 33 < 0$
$8b - 33 > 0$
$b > \frac{33}{8}$
$b > 4\cdot 1$
$b = 5$

9. Find the range of values of a, such that $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative.

'a' के किन मानों के लिये $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ सदैव ऋणात्मक रहेगा?

Ans. $a \in \left(-\infty, -\frac{1}{2}\right)$

Sol. $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$
for $x^2 - 8x + 32 = 0$
$a > 0,\ D = 64 - 4 \times 32$
$= 64 - 128$
< 0
$\therefore\ x^2 - 8x + 32 > 0$
for $ax^2 + 2(a+1)x + 9a + 4 = 0$
$a < 0,\ D < 0$
$4(a+1)^2 - 4a(9a+4) < 0$
$a^2 + 1 + 2a - 9a^2 - 4a < 0$
$-8a^2 - 2a + 1 < 0$
$8a^2 + 2a - 1 > 0$
$8a^2 + 4a - 2a - 1 > 0$
$4a(2a+1) - 1(2a+1) > 0$
$(2a+1)(4a-1) > 0$
$a \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$
But $a < 0$
$a \in \left(-\infty, -\frac{1}{2}\right)$

10. We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \le p$ is satisfied for all real x. Find the smallest integral good number.

'p' एक अच्छी संख्या कहलाता है यदि असमिका $\frac{2x^2+2x+3}{x^2+x+1} \le p$ के सभी मानों के लिए सन्तुष्ट होती है। तो न्यूनतम पूर्णांक अच्छी संख्या ज्ञात कीजिए।

Ans. 4

Sol. $\frac{2x^2+2x+3}{x^2+x+1} \le p$

for $x^2 + x + 1 = 0$

$a > 0, D < 0$

$\therefore\ x^2 + x + 1 > 0$

$\Rightarrow 2x^2 + 2x + 3 \le p(x^2 + x + 1)$

$2x^2 - px^2 + 2x - px + 3 - p \le 0$

$(2 - p)\ x^2 + (2 - p)x + (3 - p) \le 0$

$a < 0, D \le 0$

$2 - p < 0$

$-p < -2$

$p > 2$

$(2 - p)^2 - 4\ (2 - p)\ (3 - p) \le 0$

$(2 - p)\ [2 - p - 12 + 4p] \le 0$

$(2 - p)\ [3p - 10] \le 0$

$(p - 2)\ (3p - 10) \ge 0$

$p \le 2,\ p \ge \frac{10}{3}$

$p \ge \frac{10}{3}$

$p \ge 3{\cdot}3$

$p = 4$

11. Let $x^2 + y^2 + xy + 1 \ge a(x + y)\ \forall\ x, y \in R$. Find the possible integer(s) in the range of a.

माना $x^2 + y^2 + xy + 1 \ge a(x + y)\ \forall\ x, y \in R$ है। 'a' के परिसर में उपस्थित सभी सम्भावित पूर्णांक मान ज्ञात कीजिये।

Ans. –1, 0, 1

Sol. $x^2 + y^2 + xy + 1 \ge a(x + y) \quad ; \quad \forall x, y \in R$

$x^2 + xy - ax + y^2 + 1 - ay \ge 0$

$x^2 + (y - a)x + (y^2 + 1 - ay) \ge 0$

$D \le 0$

$(y - a)^2 - 4(y^2 + 1 - ay) \le 0$

$y^2 + a^2 - 2ay - 4y^2 - 4 + 4ay \leq 0$
$a^2 + 2ay - 3y^2 - 4 \leq 0$
$-3y^2 + 2ay + a^2 - 4 \leq 0$
$3y^2 - 2ay - a^2 + 4 \geq 0$
$D \leq 0$
$4a^2 - 4 \times 3(a^2 + 4) \leq 0$
$a^2 + 3a^2 - 12 \leq 0$
$4a^2 - 12 \leq 0$
$a^2 - 3 \leq 0$
$a \in \left[-\sqrt{3}, \sqrt{3}\right]$
$a = -1, 0, 1$

GENERAL EQUATION OF SECOND DEGREE IN TWO VARIABLES

12. Show that the equation $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$ for every real value of x, there is a real value of y and for every real value of y there is a real value of x.

सिद्ध कीजिये कि समीकरण $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$ में x के प्रत्येक वास्तविक मान के लिए y का एक वास्तविक मान होगा तथा y के प्रत्येक वास्तविक मान के लिए x का एक वास्तविक मान होगा।

Ans.

Sol. $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$

$a = 1, b = 2, \ h = \frac{-3}{2}, g = -1, \ f = \frac{-3}{2}, c = -35$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$$1 \times 2 \times (-35) + 2 \times \left(\frac{-3}{2}\right)(-1) \times \left(\frac{-3}{2}\right) - 1 \times \left(\frac{-3}{2}\right)^2 - 2 \times (-1)^2 - (-35) \times \left(\frac{-3}{2}\right)^2$$

$$= -70 + 3 \times \left(\frac{-3}{2}\right) - \frac{9}{4} - 2 + 35 \times \frac{9}{4}$$

$$= \frac{-70}{1} - \frac{9}{2} - \frac{9}{4} - 2 + \frac{315}{4}$$

$$= \frac{-280 - 18 - 9 - 8 + 315}{4}$$

$$= \frac{-315 + 315}{4} = 0$$

Hence proved.

13. If α, β and γ are the roots of the equation $5x^3 - qx - 1 = 0$, $(q \in R)$ then find the value of $\frac{\alpha^2-3}{\beta\gamma}+\frac{\beta^2-3}{\gamma\alpha}+\frac{\gamma^2-3}{\alpha\beta}$.

यदि α, β एवं γ समीकरण $5x^3 - qx - 1 = 0$, $(q \in R)$ के मूल है, तो $\frac{\alpha^2-3}{\beta\gamma}+\frac{\beta^2-3}{\gamma\alpha}+\frac{\gamma^2-3}{\alpha\beta}$ का मान ज्ञात कीजिये।

Ans. 3

Sol. $5x^3 - qx - 1 = 0$, $(q \in R)$

$\alpha + \beta + \gamma = 0$

$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{-q}{5}$

$\alpha\beta\gamma=\frac{1}{5}$

Now $\frac{\alpha^2-3}{\beta\gamma}+\frac{\beta^2-3}{\gamma\alpha}+\frac{\gamma^2-3}{\alpha\beta}$

$=\frac{1}{\alpha\beta\gamma}\left[\alpha^3-3\alpha+\beta^3-3\beta+\gamma^3-3\gamma\right]$

$=\frac{1}{\alpha\beta\gamma}\left[\alpha^3+\beta^3+\gamma^3-3(\alpha+\beta+\gamma)\right]$

$\because \alpha+\beta+\gamma=0 \Rightarrow \alpha^3+\beta^3+\gamma^3=3\alpha\beta\gamma$

$=\frac{1}{\alpha\beta\gamma}\left[3\alpha\beta\gamma-3(\alpha+\beta+\gamma)\right]$

$=\frac{3\alpha\beta\gamma-0}{\alpha\beta\gamma}=3$

14. Let α, β, γ are the roots of the cubic equation $x^3 + 2x^2 - x - 3 = 0$. If the absolute value of the expression $\frac{\alpha+3}{\alpha-2}+\frac{\beta+3}{\beta-2}+\frac{\gamma+3}{\gamma-2}$ can be expressed as $\frac{m}{n}$ where m and n are coprime, then find the value of (m + n).

माना α, β, γ समीकरण $x^3 + 2x^2 - x - 3 = 0$ के मूल है यदि व्यंजक $\frac{\alpha+3}{\alpha-2}+\frac{\beta+3}{\beta-2}+\frac{\gamma+3}{\gamma-2}$ का निरपेक्ष मान $\frac{m}{n}$ (जहां m तथा n सापेक्ष अभाज्य संख्या है) द्वारा प्रदर्शित किया जाता है तब (m + n) का मान ज्ञात कीजिये।

Ans. 73

Sol. Equation : $x^3 + 2x^2 - x - 3 = 0$

roots = α, β, γ

$\alpha + \beta + \gamma = -2$
$\alpha\beta + \beta\gamma + \gamma\alpha = -1$
$\alpha\beta\gamma = -3$

Let $X = \frac{x+3}{x-2}$
$xX - 2X = x + 3$
$xX - x = 3 + 2x$
$x(X - 1) = 3 + 2x$

Put $x = \frac{3+2X}{X-1}$

$\left(\frac{3+2X}{X-1}\right)^3 + 2\left(\frac{3+2X}{X-1}\right)^2 - \left(\frac{3+2X}{X-1}\right) - 3 = 0$
$(3 + 2X)^3 + 2(3 + 2X)^2(X - 1) - (3 + 2X)(X - 1)^2 - 3(X - 1)^3 = 0$
$27 + 8X^3 + 18X(3 + 2X) + (2X - 2)(9 + 4X^2 + 12X) - (3 + 2X)\cdot(X^2 + 1 - 2X) - 3(X^3 - 1 - 3X(X - 1)) = 0$
$27 + 8X^3 + 54X + 36X^2 + 18X + 8X^3 + 2X^2 - 18 - 8X^2 - 24X - 3X^2 - 3 + 6X - 2X^3 - 2X + 4X^2 - 3X^3 + 3 - 9X^2 - 9X = 0$
$11X^3 + 62X^2 + 43X + 9 = 0$
$11x^3 + 62x^2 + 43x + 9 = 0$

$\frac{\alpha+3}{\alpha-2}, \frac{\beta+3}{\beta-2}, \frac{\gamma+3}{\gamma-2}$ are the roots

$S_1 = \frac{-62}{11}$

$|S_1| = \frac{62}{11} = \frac{m}{n} \Rightarrow m + n = 62 + 11 = 73$

15. Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ and $\frac{\gamma}{\gamma-2}$. Hence or otherwise find the value of $(\alpha-2)(\beta-2)(\gamma-2)$.

माना α, β एवं γ तीन घातीय समीकरण $x^3 - 3x^2 + 1 = 0$ के मूल है। वह तीन घातीय समीकरण निकालिये जिसके मूल $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ तथा $\frac{\gamma}{\gamma-2}$ है एवं $(\alpha - 2)(\beta - 2)(\gamma - 2)$ का मान भी निकालिए।

Ans. $3y^3 - 9y^2 - 3y + 1 = 0$; $(\alpha - 2)(\beta - 2)(\gamma - 2) = 3$

Sol. $x^3 - 3x^2 + 1 = 0$
α, β, γ are the roots.
$x = \alpha$
cubic in y

$y = \frac{\alpha}{\alpha - 2} = \frac{x}{x-2}$

$xy - 2y = x$

$x(y - 1) = 2y$

Put $x = \frac{2y}{y-1}$

$\left(\frac{2y}{y-1}\right)^3 - 3\left(\frac{2y}{y-1}\right)^2 + 1 = 0$

$\frac{8y^3}{(y-1)^3} - \frac{3 \times 4y^2}{(y-1)^2} + 1 = 0$

$8y^3 - 12y^2 \cdot (y - 1) + (y - 1)^3 = 0$

$8y^3 - 12y^3 + 12y^2 + y^3 - 1 - 3y^2 + 3y = 0$

$9y^3 - 12y^3 + 9y^2 + 3y - 1 = 0$

$-3y^3 + 9y^2 + 3y - 1 = 0$

$3y^3 - 9y^2 - 3y + 1 = 0$

cubic in Z

$z = \alpha - 2$

$z = x - 2$

Put $x = (z + 2)$

$(z + 2)^3 - 3(z + 2)^2 + 1 = 0$

$z^3 + 8 + 6z(z + 2) - 3(z^2 + 4 + 4z) + 1 = 0$

$z^3 + 8 + 6z^2 + 12z - 3z^2 - 12 - 12z + 1 = 0$

$z^3 + 3z^2 - 3 = 0$

$\therefore$ product of roots $= (\alpha - 2)(\beta - 2)(\gamma - 2) = 3$

16. Two roots of a biquadratic $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ have their product equal to (-32). Find the value of k.

समीकरण $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ के दो मूलों का गुणनफल (-32) के बराबर है तो k का मान निकालिए।

Ans. K = 86

Sol. $x^4 - 18x^3 + kx^2 + 200x - 1984$

$\alpha, \beta, \gamma, \delta$ are the roots.

$\alpha + \beta + \gamma + \delta = 18$

$\Sigma\alpha\beta = k$

$\Sigma\alpha\beta\gamma = -200$

$\Sigma\alpha\beta\gamma\delta = -1984$

$\because \alpha\beta = -32$

$\Rightarrow -32 \times \gamma\delta = -1984$

$\gamma\delta = \frac{1984}{32}$

$\gamma\delta = 62$

$\because\ \alpha\beta\gamma + \alpha\beta\delta + \beta\gamma\delta + \alpha\gamma\delta = -200$

$-32\gamma - 32\delta + 62\beta + 62\alpha = -200$

$31\alpha + 31\beta - 16\gamma - 16\ \delta = -100$

$31(\alpha + \beta) - 16\ (\gamma + \delta) = -100$

put $\alpha + \beta = A,\ \gamma + \delta = B$

$31A - 16B = 100$ ...(i)

and $(\alpha + \beta) + (\gamma + \delta) = 18$

$A + B = 18$

$A = 18 - B$

put in (i)

$31\ (18 - B) - 16\ B = -100$

$558 - 31B - 16\ B = -100$

$-47B = -100 - 558$

$-47B = 658$

$B = \dfrac{658}{47} = 14$

$B = 14,\ A = 4$

$\therefore\ \alpha + \beta = 4,\ \gamma + \delta = 14$

$\because \Sigma\alpha\beta = k$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = k$

$\alpha(\beta + \gamma + \delta) + \beta(\gamma + \delta) + \gamma\delta = k$

$\alpha\cdot(\beta + 14) + \beta(14) + 62 = k$

$k = \alpha\beta + 14\ (\alpha + \beta) + 62$

$= -32 + 14 \times 4 + 62$

$= 30 + 56$

$= 86$

LOCATION OF ROOTS

17. Find the complete set of real values of 'a' for which both roots of the quadratic equation $(a^2 - 6a + 5)\ x^2 - \sqrt{a^2 + 2a}\ \ x + (6a - a^2 - 8) = 0$ lie on either side of the origin.

'a' के मानों का सम्पूर्ण समुच्चय जिसके लिए द्विघात समीकरण $(a^2 - 6a + 5)\ x^2 - \sqrt{a^2 + 2a}\ \ x + (6a - a^2 - 8) = 0$, के सभी मूल, मूल बिन्दु के दोनों ओर स्थित है।

Ans. $(-\infty, -2] \cup [\ 0, 1) \cup (2, 4) \cup (5, \infty)$

Sol. $\left(a^2 - 6a + 5\right)x^2 - \sqrt{a^2 + 2a}x + \left(6a - a^2 - 8\right) = 0$

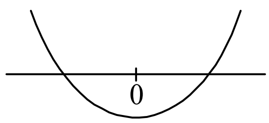
$a > 0,\ D > 0,\ a\cdot f(0) < 0$

$a^2 - 6a + 5 > 0$

$a^2 - 5a - a + 5 > 0$

$a(a - 5) - 1\ (a - 5) > 0$

$(a - 1)\ (a - 5) > 0$



$a \in (-\infty, 1) \cup (5, \infty)$

$(a^2 + 2a) - 4(a^2 - 6a + 5)\cdot(6a - a^2 - 8) > 0$
$a^2 + 2a - 4\,(6a^3 - a^4 - 8a^2 - 36a^2 + 6a^3 + 48a + 30a - 5a^2 - 40) > 0$
$a^2 + 2a - 24a^3 + 4a^4 + 32a^2 + 144a^2 - 24a^3 - 192a - 120a + 20a^2 + 160 > 0$
$4a^4 - 48a^3 + 197a^2 - 310a + 160 > 0$
$a \in R$
$a\cdot f(0) < 0$
$(a^2 - 6a + 5)\cdot(6a - a^2 - 8) < 0$
$(a - 1)\,(a - 5)\,(a - 2)\,(a - 4) > 0$
$a \in (-\infty, 1) \cup (2, 4) \cup (5, \infty)$
And $a^2 + 2a \geq 0$
$a\,(a + 2) \geq 0$
$\because$ roots are real.
$a \in (-\infty, -2] \cup [0, +\infty)$
(i) $\cap$ (ii) $\cap$ (iii)
$a \in (-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty)$

18. If α, β are the roots of the equation, $x^2 - 2x - a^2 + 1 = 0$ and γ, δ are the roots of the equation, $x^2 - 2\,(a + 1)\,x + a\,(a - 1) = 0$ such that $\alpha, \beta \in (\gamma, \delta)$ then find the values of 'a'.
यदि α, β समीकरण $x^2 - 2x - a^2 + 1 = 0$ के तथा γ, δ समीकरण $x^2 - 2\,(a + 1)\,x + a\,(a - 1) = 0$ के मूल इस प्रकार है की $\alpha, \beta \in (\gamma, \delta)$ तो 'a' के मान निकालिये।

Ans. $a \in \left(-\frac{1}{4}, 1\right)$

Sol. $x^2 - 2x - a^2 + 1 = 0$ $\qquad$ $x^2 - 2(a + 1)x + a(a - 1) = 0$
α, β are the roots. $\qquad$ γ, δ are the roots.
$\because \alpha, \beta \in (\gamma, \delta)$
from graph :
$a\cdot f(\alpha) < 0$, $\qquad a\cdot f(\beta) < 0 \qquad$ for $f(x) = x^2 - 2(a + 1)x + a(a - 1)$
from Equation (i)

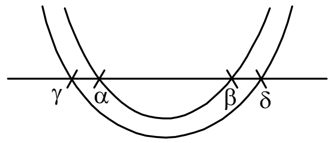
$$x = \frac{2 \pm \sqrt{4 - 4(-a^2 + 1)}}{2}$$

$$x = \frac{2 \pm \sqrt{4 + 4a^2 - 4}}{2}$$

$$x = \frac{2 \pm 2a}{2}$$

$x = \underset{\alpha}{1 + a}, \ \underset{\beta}{1 - a}$
for 2nd equation
$a\cdot f(\alpha) < 0$
$a\cdot f(1 + a) < 0$

$1\cdot\{(1+a)^2-2(a+1)\cdot(1+a)+a(a-1)\}<0$
$\{-(a+1)^2+a^2-a\}<0$
$-a^2-1-2a+a^2-a<0$
$-3a-1<0$
$3a+1>0$



$a>\frac{-1}{3}$
and $a\cdot f(\beta)<0$
$1\cdot f(1-a)<0$
$\{(1-a)^2-2(a+1)(1-a)+(a-1)\}<0$
$1+a^2-2a+2(a^2-1)+a^2-a<0$
$1+a^2-2a+2a^2-2+a^2-a<0$
$4a^2-3a-1<0$
$4a^2-4a+a-1<0$
$4a(a-1)+1(a-1)<0$
$(a-1)(4a+1)<0$
$-\frac{1}{4}<a<1$
Final answer = (i) $\cap$ (ii)
$a\in\left(-\frac{1}{4},1\right)$

19. Find all the values of the parameter 'a' for which both roots of the quadratic equation $x^2-ax+2=0$ belong to the interval (0, 3).

प्राचल 'a' के सभी मान बताइये जिनके लिये द्विघात समीकरण $x^2-ax+2=0$ के दोनों मूल अन्तराल (0, 3) में स्थित है।

Ans. $2\sqrt{2}\le a<\frac{11}{3}$

Sol. $x^2-ax+2=0$
a, b are the roots.
(i) $D\ge 0$
$a^2-8\ge 0$
$a\le -2\sqrt{2}$, $a\ge 2\sqrt{2}$
(ii) $a\cdot f(0)>0$
$2>0$
(iii) $a\cdot f(3)>0$
$9-3a+2>0$
$3a-11<0$
$a<\frac{11}{3}$

(iv) $0<\frac{-b}{2a}<3$

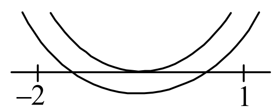
$$0 < \frac{a}{2} < 3$$
$0 < a < 6$
$a \in (0, 6)$
(i) $\cap$ (ii) $\cap$ (iii) $\cap$ (iv)
$$a \in \left[2\sqrt{2}, \frac{11}{3}\right)$$

20. At what values of 'a' do all the zeroes of the function, f'x) = $(a-2)x^2 + 2ax + a + 3$ lie on the interval $(-2, 1)$?

'a' किस किन मानों के लिये फलन f'x) = $(a-2)x^2 + 2ax + a + 3$ के सभी शून्य अन्तराल $(-2, 1)$ में स्थित होगें?

Ans. $\left(-\infty, -\frac{1}{4}\right) \cup \{2\} \cup (5, 6]$

Sol. $f(x) = (a-2)x^2 + 2ax + (a+3)$



(i) $D \geq 0$
$4a^2 - 4(a-2)(a+3) \geq 0$
$a^2 - (a^2 + a - 6) \geq 0$
$a^2 - a^2 - a + 6 \geq 0$
$a - 6 \leq 0$
$a \leq 6$

(ii) $-2 < \frac{-b}{2a} < 1$

$$-2 < \frac{-2a}{2(a-2)} < 1$$

for $\frac{-a}{a-2} > -2$

$$\frac{-a}{a-2} + 2 > 0$$

$$\frac{-a + 2a - 4}{a-2} > 0$$

$$\frac{a-4}{a-2} > 0$$

$a < 2, a > 4$

for $\frac{-a}{a-2} < 1$

$$\frac{-a}{a-2} - 1 < 0$$

$$\frac{-a-a+2}{a-2}<0$$

$$\frac{-2a+2}{a-2}<0$$

$$\frac{2(a-1)}{a-2}>0$$

$$\frac{a-1}{a-2}>0$$

$\Rightarrow a < 1, a > 4$

(iii) $a \cdot f(-2) > 0$

$(a-2)\ \{(a-2)4 - 4a + a + 3\} > 0$

$(a-2)\ \{4a - 8 - 4a + a + 3\} > 0$

$(a-2)\ (a-5) > 0$

$a < 2, a > 5$

and (iv) $a \cdot f(1) > 0$

$(a-2)\ \{a - 2 + 2a + a\ + 3\} > 0$

$(a-2)\ \{4a + 1\} > 0$

$$a < -\frac{1}{4}, a > 2$$

(i), (ii), (iii) & (iv)

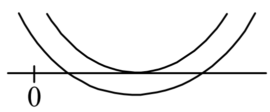
$$a \in \left(-\infty, \frac{-1}{4}\right) \cup \{2\} \cup (5, 6]$$

21. Find the values of K so that the quadratic equation $x^2 + 2(K-1)x + K + 5 = 0$ has atleast one positive root.

K के किन मानों के लिये द्विघात समीकरण $x^2 + 2(K-1)x + K + 5 = 0$ का कम से कम एक धनात्मक मूल है।

Ans. $K \le -1$

Sol. $x^2 + 2\ (k-1)x + k + 5 = 0$



(i) $D \ge 0$

$4(k-1)^2 - 4(k+5) \ge 0$

$k^2 + 1 - 2k - k - 5 \ge 0$

$k^2 - 3k - 4 \ge 0$

$(k+1)\ (k-4) \ge 0$

$$\frac{-b}{2a} > 0$$

$$\frac{-2(k-1)}{2} > 0$$

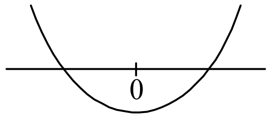
$k < 1$

$k \in (-\infty, 1)$

$f(0) > 0$

$k + 5 > 0$

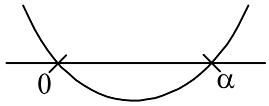
$k > -5$
$k \in (-5, \infty)$
Intersection value
$k \in (-5, -1]$
(ii) $f(0) < 0$
$k + 5 < 0$
$k \in (-\infty, -5)$



(iii) $f(0) = 0$; $-\dfrac{b}{2a} > 0$; $D > 0$

$k = -5$; $\dfrac{-2(k-1)}{2} > 0$; $k < -1,\ k > 4$

$k - 1 < 0$
$k < 1$



Intersection value
$\Rightarrow \quad k = -5$
Final answer $k \le -1$

22. If all the solutions of the inequality $x^2 - 6ax + 5a^2 \le 0$ are also the solutions of inequality $x^2 - 14x + 40 \le 0$ then find the number of possible integral values of a.

यदि असमिका $x^2 - 6ax + 5a^2 \le 0$ के सभी हल असमिका $x^2 - 14x + 40 \le 0$ के भी हल है तो a के सम्भावित पूर्णांक मानों की संख्या बताओ।

Ans. 0

Sol. $x^2 - 14x + 40 \le 0$
$x^2 - 4x - 10x + 40 \le 0$
$x(x-4) - 10(x-4) \le 0$
$(x-4)(x-10) \le 0$
$x \in [4, 10]$
for $x^2 - 6ax + 5a^2 \le 0$
$x^2 - 5ax - ax + 5a^2 \le 0$
$x(x-5a) - a(x-5a) \le 0$
$(x-5a)(x-a) \le 0$
Case–I :
if $a > 0$
$x \in [a, 5a]$
$\because\ a \ge 4,\ 5a \ge 10$
$a \ge 4,\ a \ge 2$
Not possible.
Case–II : $a < 0$
$x \in [5a, a]$
$\therefore\ x \in [4, 10]$
$\therefore$ No solution
$\therefore$ There is no integral value for a.

23. The smallest value of k, for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is______.

k का वह न्यूनतम मान जिसके लिये समीकरण $x^2 - 8kx + 16(k^2 - k + 1) = 0$ के दोनों मूल वास्तविक एवं भिन्न है तथा कम से कम 4 के बराबर हैं, ______ है

Ans. 2

Sol. $x^2 - 8kx + 16(k^2 - k + 1) = 0;$

$D > 0$;	$\frac{-b}{2a} > 4$;	$f(4) \geq 0$
$64k^2 - 64(k^2 - k + 1)$;	$\frac{8k}{2} > 4$;	$16 - 32k + 16k^2 - 16k + 16 \geq 0$
$k - 1 > 0$;	$k > 1$;	$16k^2 - 48k + 32 \geq 0$
$k > 1$;		$k \leq 1, k \geq 2$

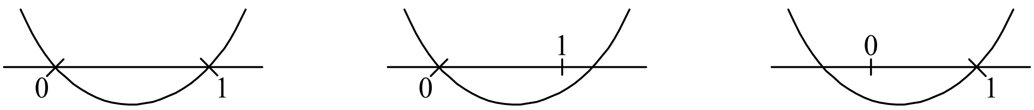
$\Rightarrow$ Minimum value of $k = 2$

24. Largest integral value of 'a' for which the expression $y = x^2 + ax + (a - 3)$ is non positive $\forall\ x \in [0, 1]$ is______.

यदि व्यंजक $y = x^2 + ax + (a - 3)$ हमेशा या तो शून्य है या ऋणात्मक तो जहाँ $\forall\ x \in [0, 1]$ तो a का अधिकतम पूर्णांक में मान होगा______.

Ans. 1

Sol.



$y = ax^2 + ax + (a - 3),\ \forall x \in [0,1]$

Let $f(x) = x^2 + ax + (a - 3)$

$\because\ f(0) \leq 0$ $\qquad$ $f(1) \leq 0$

$a - 3 \leq 0$ $\qquad$ $1 + a + a - 3 \leq 0$

$a \leq 3$ $\qquad$ $2a - 2 \leq 0$

$a \leq 1$

$\therefore\ a_{max} = 1$

25. If the equation $x^4 - ax^3 + 11x^2 - ax + 1 = 0$ has four distinct positive roots then the range of a is (m, M). Find the value of m + 2M.

यदि समीकरण $x^4 - ax^3 + 11x^2 - ax + 1 = 0$ के चार भिन्न धनात्मक मूल है तो a का परिसर (m, M) है। m + 2M का मान ज्ञात कीजिए।

Ans. 19

Sol. $x^4 - ax^3 + 11x^2 - ax + 1 = 0$

divide by x^2

$$x^2 - ax + 11 - \frac{a}{x} + \frac{1}{x^2} = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) - a\left(x + \frac{1}{x}\right) + 11 = 0$$

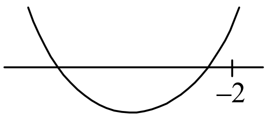
put $x + \frac{1}{x} = t$

$$x^2 + \frac{1}{x^2} = t^2 - 2$$

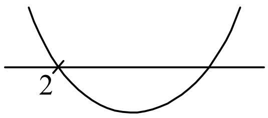
$t^2 - 2 - at + 11 = 0$

$t^2 - at + 9 = 0$

$\because x + \frac{1}{x} \le -2$, $\quad x + \frac{1}{x} \ge 2$



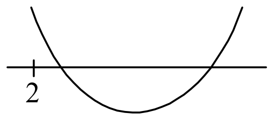
Impossible



$x + \frac{1}{x} = 2$

$x^2 - 2x + 1 = 0$

$x = 1$ not possible



$x + \frac{1}{x} = 2$

$x^2 - 2x + 1 = 0$

$(x - 1)^2 = 0$ {two equal roots}

$x = 1$ Not possible

$t^2 - at + 9 = 0$

t_1, t_2 are the roots.

(i) $D > 0$

$a^2 - 36 > 0$

$(a + 6)(a - 6) > 0$

$a \in (-\infty, -6) \cup (6, \infty)$

(ii) $\frac{-b}{2a} > 2$

$\frac{a}{2} > 2$

$a > 4$

$a \in (4, \infty)$

(iii) $a \cdot f(2) > 0$

$1 \times (4 - 2a + 9) > 0$

$2a - 13 < 0$

$a < \frac{13}{2}$

$a \in \left(-\infty, \frac{13}{2}\right)$

(i), (ii), (iii)

$a \in \left(6, \frac{13}{2}\right)$

$m = 6,\ M = \frac{13}{2}$

$\Rightarrow m + 2M = 6 + 2 \cdot \frac{13}{2} = 19$

MISCELLANEOUS

26. When $y^2 + my + 2$ is divided by $(y - 1)$ then the quotient is f (y) and the remainder is R_1. When $y^2 + my + 2$ is divided by $(y + 1)$ then quotient is g (y) and the remainder is R_2. If $R_1 = R_2$ then find the value of m.

जब $y^2 + my + 2$ को $(y - 1)$ से भाग दिया जाता है तो भागफल f (y) मिलता है तथा शेषफल R_1 है। जब $y^2 + my + 2$ का $(y + 1)$ से भाग दिया जाता है तो भागफल g (y) है तथा शेषफल R_2 होगा। यदि $R_1 = R_2$ हो तो m का मान ज्ञात कीजिए।

Ans. 0

Sol. $y^2 + my + 2 = (y - 1) f(y) + R_1$ ...(i)
$\&\ y^2 + my + 2 = (y + 1) g(y) + R_2$ ...(ii)
in Equation (i), put y = 1
$1 + m + 2 = R^1$
$R_1 = m + 3$
in Equation (ii), put y = 1
$1 - m + 2 = R_2$
$R_2 = 3 - m$
$\because R_1 = R_2$
$m + 3 = 3 - m$
$2m = 0$
$m = 0$

27. Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. Find the unique pair of real numbers (x, y) that satisfy $P(x) \cdot Q(y) = 28$.

माना $P(x) = 4x^2 + 6x + 4$ तथा $Q(y) = 4y^2 - 12y + 25$ है। वास्तविक संख्याओं x तथा y का एकल युग्म (x, y) बताइये तो $P(x) \cdot Q(y) = 28$ को संतुष्ट करता है।

Ans. $\left(-\frac{3}{4}, \frac{3}{2}\right)$

Sol. $P(x) = 4x^2 + 6x + 4$
$\because\ a > 0,\ P(x)_{max} = \infty$
for minimum of P(x)
$\left(\frac{-b}{2a}, \frac{-D}{4a}\right) = \left(\frac{-6}{8}, 4 - \frac{36}{4 \times 4}\right)$
$= \left(\frac{-3}{4}, \frac{7}{4}\right)$

$Q(x) = 4y^2 - 12y + 25$
$a > 0,\ Q(y)_{max} = \infty$
for minimum of Q(y)
$\left(\frac{-b}{2a}, \frac{-D}{4a}\right) = \left(\frac{12}{8}, 25 - \frac{144}{4 \times 4}\right)$
$= \left(\frac{3}{2}, 16\right)$
$= (y, Q(y))$

$\Rightarrow P(x) \cdot P(y) = \frac{7}{4} \times 16 = 28$

$(x, y) = \left(\frac{-3}{4}, \frac{3}{2}\right)$

28. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$.

समीकरण $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$ के वास्तविक मूलो का गुणनफल ज्ञात कीजिये।

Ans. 20

Sol. $x^2 + 18x + 30 = 2\sqrt{x + 18 + 45}$
Put $t = x^2 + 18x + 30$
$\therefore t = 2\sqrt{t + 15}$
$t^2 = 4(t + 15)$
$t^2 - 4t - 60 = 0$
$(t - 10)(t + 6) = 0$
$t = 10, -6$
$\therefore\ x^2 + 8x + 30 = 10$
$x^2 + 8x + 20 = 10$
$\therefore$ product of roots = 20

29. Find all real numbers x such that, $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$.

x के वास्तविक मान ज्ञात कीजिए जिनके लिये $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$ हो।

Ans. $x = \frac{\sqrt{5}+1}{2}$

Sol. $\left(x - \frac{1}{x}\right)^{\frac{1}{2}} + \left(1 - \frac{1}{x}\right)^{\frac{1}{2}} = x$

$x - \frac{1}{x} + 1 - \frac{1}{x} + 2\sqrt{\left(x - \frac{1}{x}\right)\left(1 - \frac{1}{x}\right)} = x^2$

$x - \frac{2}{x} + 1 + 2\sqrt{x - 1 - \frac{1}{x} + \frac{1}{x^2}} = x^2$

$2\sqrt{\frac{x^3 - x^2 - x + 1}{x^2}} = x^2 - x + \frac{2}{x} - 1$

$\frac{2\sqrt{x^3 - x^2 - x + 1}}{x} = \frac{\left(x^3 - x^2 - x + 2\right)}{x}; x \neq 0$

$4(x^3 - x^2 - x + 1) = (x^3 - x^2 - x + 2)^2$

put $x^2 - x^2 - x + 1 = t$

$4t = (t + 1)^2$

$t^2 + 1 + 2t - 4t = 0$

$(t - 1)^2 = 0$

$t = 1$

$\therefore\ x^3 - x^2 - x + 1 = 1$

$x\,(x^2 - x - 1) = 0$

$x \neq 0$

$x^2 - x - 1 = 0$

$x = \frac{1 \pm \sqrt{5}}{2}$

$x = \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}$

$\therefore x = \frac{\left(1+\sqrt{5}\right)}{2}$

30. Solve : $(x + 9)\,(x - 3)\,(x - 7)\,(x + 5) = 385$

हल करो : $(x + 9)\,(x - 3)\,(x - 7)\,(x + 5) = 385$

Ans. $-4, 2$ and $-1 \pm \sqrt{71}$

Sol. $(x + 9)\,(x - 3)\,(x - 7)\,(x + 5) = 385$

$(x + 9)\,(x - 7)\,(x - 3)\,(x + 5) = 385$

$(x^2 + 2x - 63)(x^2 + 2x - 15) = 385$
put $x^2 + 2x = t$
$(t - 63)(t - 15) = 385$
$t^2 - 15t - 63t + 945 - 385 = 0$
$t^2 - 78t + 560 = 0$
$t^2 - 70t - 8t + 560 = 0$
$(t - 70)(t - 8) = 0$
$t = 70$ and 8

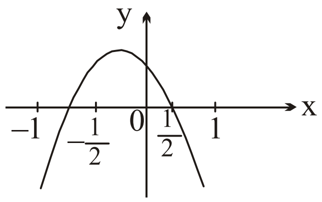
for $t = 70$
$x^2 + 2x - 70 = 0$
$$x = \frac{-2 \pm \sqrt{4 + 4 \times 70}}{2}$$
$x = -1 \pm \sqrt{71}$

for $t = 8$
$x^2 + 2x - 8 = 0$
$x^2 + 4x - 2x - 8 = 0$
$x(x + 4) - 2(x + 4) = 0$
$(x + 4)(x - 2) = 0$
$x = -4, 2$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 31 to 32

Let $f(x) = ax^2 + bx + c$ be a quadratic expression and $y = f(x)$ has graph as shown in figure
यदि $f(x) = ax^2 + bx + c$ एक द्विघात व्यंजक है। एवं $y = f(x)$ का ग्राफ चित्रानुसार है।



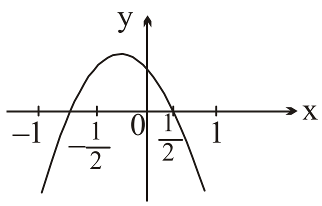
31. Which of the following is false?

निम्न में से कौनसा विकल्प असत्य है?

(A) $ab > 0$ (B) $abc < 0$ (C) $ac < 0$ (D) $bc < 0$

Ans. B

Sol. $f(x) = ax^2 + bx + c$
$a < 0$
$$-\frac{b}{2a} < 0$$
$$\frac{b}{2a} > 0$$
but $a < 0 \Rightarrow b < 0$ and $c > 0$
(A) Now $a \cdot b = (-)(-) = +ve$
$\Rightarrow ab > 0$



(B) abc = (–) (–) (+) = +ve
abc> 0
(C) ac < 0
= (–) (+)
= (–)
(D) bc = (–) (+) = (–)
bc < 0
$\therefore$ option B is false.

32. Which of the following is true ?

निम्न में से कौनसा विकल्प सत्य है?

(A) $a + b + c > 0$ (B) $a - b + c > 0$ (C) $a + 3b + 9c < 0$ (D) $a - 3b + 9c > 0$

Ans. D

Sol. $f(x) = ax^2 + bx + c$
(i) $f(1) < 0$
$a + b + c < 0$
(ii) $f(-1) < 0$
$a - b + c < 0$
(iii) $f\left(-\frac{1}{3}\right) > 0$

$\frac{a}{9} - \frac{b}{3} + c > 0$
$a - 3b + 9c > 0$

Paragraph for Question nos. 33 to 35

Consider a rational function $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ and a quadratic function

$g(x) = x^2 - (m + 1)x + m - 1$, where m is a parameter.

माना एक परिमेय फलन $f(x) = \frac{x^2 + 3x + 1}{x^2 + x + 1}$ है तथा एक द्विघात फलन

$g(x) = x^2 - (m + 1)x + m - 1$ है, जहाँ m प्राचल है।

33. Number of integral value(s) of 'm' so that g(x) is always positive, is :

(A) 0 (B) 1 (C) 2 (D) more than 2

g(x) सदैव धनात्मक है तो 'm' के पूर्णांक मानों की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 2 से अधिक

Ans. A

Sol. $f(x)=\dfrac{x^2+3x+1}{x^2+x+1}$ and $g(x) = x^2 - (m+1)x + m - 1$

for $g'(x)$ is always +ve.

$a > 0$, $D < 0$

$(m+1)^2 - 4(m-1) < 0$

$m^2 + 1 + 2m - 4m + 4 < 0$

$m^2 - 2m + 1 + 4 < 0$

$(m-1)^2 + 4 < 0$

No value of m.

$\therefore$ Option A

34. Number of integral values in the range of f(x), is :

f(x) के परिसर में पूर्णांक मानों की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) more than 3

Ans. C

Sol. Let $y=\dfrac{x^2+3x+1}{x^2+x+1}$

Range of y

$yx^2 + yx + y - x^2 - 3x - 1 = 0$

$(y-1)x^2 + (y-3)x + (y-1) = 0$

$D \geq 0$

$(y-3)^2 - 4(y-1)^2 \geq 0$

$y^2 + 9 - 6y - 4y^2 - 4 + 8y \geq 0$

$-3y^2 + 2y + 5 \geq 0$

$3y^2 - 2y - 5 \leq 0$

$3y^2 + 3y - 5y - 5 \leq 0$

$3y(y+1) - 5(y+1) \leq 0$

$(y+1)(3y-5) \leq 0$

$-1 \leq y \leq \dfrac{5}{3}$

$y = -1, 0, 1$ (Integral Values)

35. If both roots of $g(x) = 0$ are greater than the smallest value of the function f(x), then 'm' lies in the interval :

यदि $g(x) = 0$ के दोनों मूल f(x) के न्यूनतम मान से अधिक है, तो 'm' किस अन्तराल में होगा :

(A) $(-\infty, -2)$ (B) $\left(-\infty,-\dfrac{1}{4}\right)$ (C) $(-2, \infty)$ (D) $\left(-\dfrac{1}{2},\infty\right)$

Ans. D

Sol. $\because f(x) \in \left[-1, \frac{5}{3}\right]$

given that both roots of g(x) = 0 are greater than the smallest value of f(x).

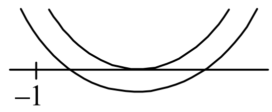
$\Rightarrow D \geq 0$

$(m + 1)^2 - 4(m - 1) \geq 0$

$m^2 + 1 + 2m - 4m + 4 \geq 0$

$(m - 1)^2 + 4 \geq 0$

$m \in R$



$\Rightarrow \frac{-b}{2a} > -1$

$\frac{(m+1)}{2} > -1$

$m + 1 > -2$

$m > -3$

$\Rightarrow f(-1) > 0$

$1 + m + 1 + m - 1 > 0$

$2m + 1 > 0$

$m > \frac{-1}{2}$

$\Rightarrow m \in \left(-\frac{1}{2}, +\infty\right)$

D is correct.

Paragraph for Question nos. 36 to 37

For a, b $\in R - \{0\}$, let $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in R$.

Also the equation $f(x) = 7x + a$ has only one real and distinct solution.

a, b $\in R - \{0\}$ के लिए, माना $f(x) = ax^2 + bx + a$, $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in R$ को संतुष्ट करता है तथा समीकरण $f(x) = 7x + a$ का केवल एक वास्तविक तथा भिन्न हल है।

36. The value of $(a + b)$ is equal to :

$(a + b)$ का मान बराबर है :

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. For a, b $\in R - \{0\}$, let $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right) \forall x \in R$.

Also the equation $f(x) = 7x + a$ has only one real and distinct solution.

$$\Rightarrow f\left(x+\frac{7}{4}\right)=f\left(\frac{7}{4}-x\right)$$

$$a\left(x+\frac{7}{4}\right)^2+b\left(x+\frac{7}{4}\right)+a=a\left(\frac{7}{4}-x\right)^2+b\left(\frac{7}{4}-x\right)+a$$

$$a\left(x^2+\frac{49}{16}+\frac{7x}{2}\right)+bx+\frac{7b}{4}=a\left(\frac{49}{16}+x^2-\frac{7}{2}x\right)+\frac{7b}{4}-bx$$

$$ax^2+\frac{49}{16}a+\frac{7ax}{2}+bx=ax^2+\frac{49}{16}a-\frac{7ax}{2}-bx$$

$7ax = -2bx$

$7a + 2b = 0$

And $f(x) = 7x + a$

$ax^2 + bx + a = 7x + a$

$ax^2 + (b - 7)x = 0$

$\because$ one root is real and distinct.

$D = 0$

$\therefore (b - 7)^2 + 4 \times a \times 0 = 0$

$b = 7$

$\Rightarrow 7a + 2 \times 7 = 0$

$7a = -14$

$a = -2$

$a + b = -2 + 7 = 5$

37. The minimum value of f(x) in $\left[0, \frac{3}{2}\right]$ is equal to :

अन्तराल $\left[0, \frac{3}{2}\right]$ में f(x) का न्यूनतम मान है :

(A) $\frac{-33}{8}$ (B) 0 (C) 4 (D) -2

Ans. D

Sol. $\because f(x) = ax^2 + bx + a$

$= -2x^2 + 7x - 2$

$$V=\left(-\frac{b}{2a},\frac{D}{4a}\right)\equiv\left(\frac{-7}{2(-2)},c-\frac{b^2}{4a}\right)\equiv\left(\frac{7}{4},-2+\frac{49}{8}\right)\equiv\left(\frac{7}{4},\frac{33}{8}\right)$$

$f(0) = 0 + 0 - 2$, $\qquad f\left(\frac{3}{2}\right)=-2\left(\frac{9}{4}\right)+7\left(\frac{3}{2}\right)-2$

$= -2$ $\qquad =-\frac{9}{2}+\frac{21}{2}-2$

$$= \frac{-9+21-4}{2} = \frac{8}{2} = 4$$

Min value of f(x) in $\left[0, \frac{3}{2}\right] = -2$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

38. The value(s) of 'p' for which the equation $ax^2 - px + ab = 0$ and $x^2 - ax - bx + ab = 0$ may have a common root, given a, b are non zero real numbers, is :

p के वह मान जिसके लिये समीकरण $ax^2 - px + ab = 0$ तथा $x^2 - ax - bx + ab = 0$ एक उभयनिष्ठ मूल रखती है। दिया गया है a, b अशून्य वास्तविक संख्या है :

(A) $a + b^2$ (B) $a^2 + b$ (C) $a(1 + b)$ (D) $b(1 + a)$

Ans. BC

Sol. Equation $ax^2 - px + ab = 0$ ...(i)

$x^2 - ax - bx + ab = 0$

from equation, $x = a, b$

$\because$ both equations may have a common root.

for $x = a$

$a^3 - ap + ab = 0$

$a(a^2 - p + b) = 0$

$a \neq 0,\ p = a^2 + b$

for $x = b$

$ab^2 - pb + ab = 0$

$b(ab - p + a) = 0$

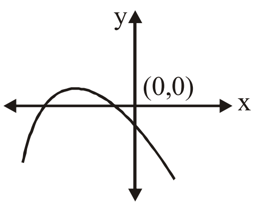
$b \neq 0$

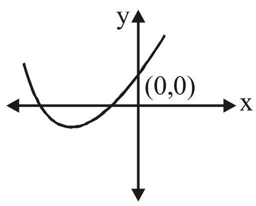
$p = a(1 + b)$

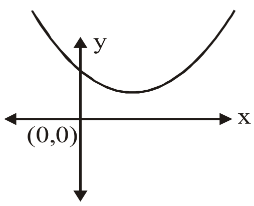
Option B, C are correct.

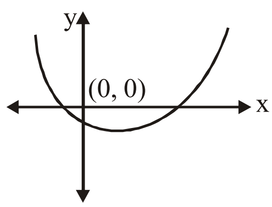
39. For which of the following graphs of quadratic expression $y = ax^2 + bx + c$, then product abc is positive?

द्विघात व्यंजक $y = ax^2 + bx + c$ के निम्न में से कौनसे आरेखों के लिए गुणन abc धनात्मक है?

(A) 

(B) 

(C) 

(D) 

Ans. BD

Sol. (A) $a < 0$

$-\frac{b}{2a} < 0$

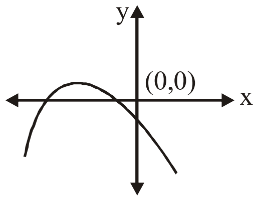
$\frac{b}{2a} > 0$

but $a < 0$

$b < 0$

and $c < 0$

$\therefore\ a \cdot b \cdot c = (-ve)\ (-ve)\ (-ve) = -ve$



(B) $a > 0$

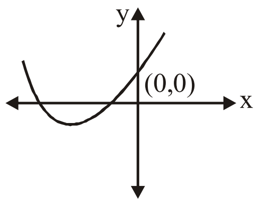
$-\frac{b}{2a} < 0$

$\frac{b}{2a} > 0$

but $a > 0 \Rightarrow b > 0$

and $c > 0$

Now abc = (+) (+) (+) = +ve



(C) $a > 0$

$-\frac{b}{2a} > 0$

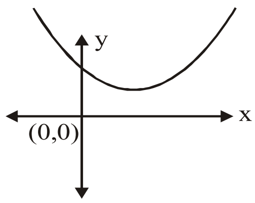
$\frac{b}{2a} < 0$

but $a > 0 \Rightarrow b < 0$

$c > 0$

Now abc = (+) (–) (+)

= –ve



(D) $a > 0$

$\frac{-b}{2a} > 0$

$\frac{b}{2a} < 0$

but $a > 0$ then $b < 0$

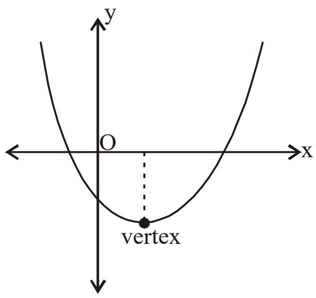
and $c < 0$

Now abc = (+) (–) (–)

= +ve

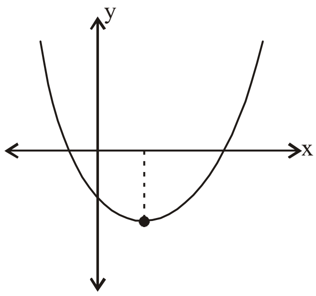
40. Graph of $y = ax^2 + bx + c = 0$ is given adjacently. What conclusions can be drawn from this graph :

फलन $y = ax^2 + bx + c = 0$ का रेखाचित्र दर्शाया गया है, तब :



(A) $a > 0$ (B) $b < 0$ (C) $c < 0$ (D) $b^2 - 4ac > 0$

Ans. ABCD



Sol. $a > 0$

$$-\frac{b}{2a} > 0$$

$$\frac{b}{2a} < 0$$

but $a > 0$ then $b < 0$

and $c < 0$, $D > 0$

$b^2 - 4ac > 0$

All options are correct.

41. The values of a for which $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ does not have a real solution is :

a के मान, जिसके लिए समीकरण $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ का कोई वास्तविक हल नहीं है, होगा :

(A) –10 (B) 12 (C) 5 (D) –30

Ans. BCD

Sol. $$\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$$

$$\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x+4)} + \frac{a}{30} = 0, \qquad x \neq 1, 2$$

$30(x - 3) + a(x + 4) = 0$

$30x - 90 + ax + 4a = 0$

$x(30 + a) = 90 - 4a$

$$x = \frac{90 - 4a}{30 + a}$$

for $a = -30$, No solution

for $x = 1$

$1 = \dfrac{90-4a}{30+a}$

$30 + a = 90 - 4a$

$5a = 60$

$a = 12$

for $x = 2$

$2 = \dfrac{90-4a}{30+a}$

$60 + 2a = 90 - 4a$

$6a = 30$

$a = 5$

for $a = 5, 12, -30$

No solution.

42. If α, β are the roots of $4x^2 - 16x + c = 0$, $c > 0$ such that $1 < \alpha < 2 < \beta < 3$, then value of 'c' can be :

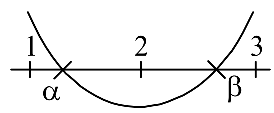
यदि α, β समीकरण $4x^2 - 16x + c = 0$, $c > 0$ के मूल है इस प्रकार कि $1 < \alpha < 2 < \beta < 3$, तो 'c' का मान हो सकता है :

(A) 11 (B) 12 (C) 13 (D) 14

Ans. CD

Sol. $4x^2 - 16x + c = 0$, $c > 0$

$\alpha < 2 < \beta < 3$



$f(1) > 0$, $f(3) > 0$

$4 - 16 + c > 0$, $36 - 48 + c > 0$

$c > 12$ $-12 + c > 0$

$c > 12$

$f(2) < 0 \Rightarrow 16 - 32 + c < 0$

$-16 + c < 0$

$c < 16$

$\because c \in (12, 16)$

Options C and D are correct.

43. If the quadratic polynomial $P(x) = (p - 3)x^2 - 2px + 3p - 6$ ranges from $[0, \infty)$ for every $x \in R$, then the value of p can be :

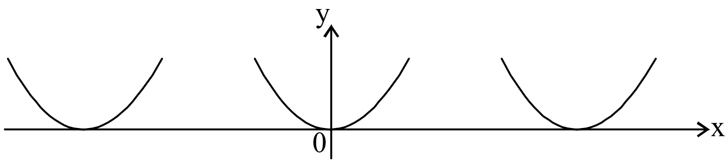
यदि द्विघात बहुपद $P(x) = (p - 3)x^2 - 2px + 3p - 6$ का परिसर $[0, \infty)$ $x \in R$ के लिए है, तो p का मान निम्न में से कौनसा हो सकता है :

(A) $\dfrac{3}{2}$ (B) 4 (C) 6 (D) 7

Ans. CD

Sol. $P(x) = (p - 3)x^2 - 2px + 3p - 6$

$\because P(x) \in [0, \infty)$



$\Rightarrow a > 0$

$P - 3 > 0 \qquad P > 3$

$\Rightarrow D \le 0$

$4p^2 - 4(p-3)(3p-6) \le 0$

$p^2 - (3p^2 - 6p - 9p + 18) \le 0$

$p^2 - 3p^2 + 15p - 18 \le 0$

$-2p^2 + 15p - 18 \le 0$

$2p^2 - 15p + 18 \ge 0$

$2p^2 - 12p - 3p + 18 \ge 0$

$2p(p-6) - 3(p-6) \ge 0$

$(p-6)(2p-3) \ge 0$

$p \ge 6,\ p \le \frac{3}{2}$

$\Rightarrow p \ge 6$

$\therefore$ Correct option C, D.

44. If $f(x) = 2x^2 + kx + 50$, $k \ge 20$ and magnitude of difference of the roots of the equation $f(x) = 0$ is 8, then which of the following can be true :

(A) minimum value of $f(x)$ is –32 (B) minimum value of $f(x)$ is –64

(C) graph of $f(x)$ cuts positive y-axis (D) graph of $f(x)$ cuts negative y-axis

यदि $f(x) = 2x^2 + kx + 50$, $k \ge 20$ तथा समीकरण $f(x) = 0$ के मूलों के अन्तर का मापांक 8 हो, तो निम्न में से कौनसा कथन सत्य हो सकता है :

(A) $f(x)$ का न्यूनतम मान –32 है (B) $f(x)$ का न्यूनतम मान –64 है।

(C) $f(x)$ का आरेख धनात्मक y-अक्ष को काटता है (D) $f(x)$ का आरेख ऋणात्मक y-अक्ष पर काटता है

Ans. AC

Sol. $f(x) = 2x^2 + kx + 50$, $k \ge 20$

$$|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right|$$

$$\left|\frac{\sqrt{D}}{a}\right| = 8$$

$\sqrt{D} = 8 \times 2$

$D = 64 \times 4 = 256$

$$\frac{-D}{4a} = \frac{-256}{4 \times 2} = -32$$

$f(0) = 50 = +ve$

$\therefore$ A, C are correct.

45. Let α and β ($\alpha > \beta$) be roots of the quadratic equation $ax^2 + bx + c = 0$. If $0 < a < c$ and $a + c = b$, then which of the following statement(s) is/are correct?

(A) Larger root is 1 (B) Larger root is – 1 (C) Smaller root is $\frac{c}{a}$ (D) b is greater than zero

माना α तथा β ($\alpha > \beta$) वर्ग समीकरण $ax^2 + bx + c = 0$ के मूल है। यदि $0 < a < c$ तथा $a + c = b$, तो निम्न में से कौनसा कथन सत्य है/हैं?

(A) बड़ा मूल 1 है (B) बड़ा मूल – 1 है (C) छोटा मूल $\frac{c}{a}$ है (D) b शून्य से बड़ा है

Ans. BD

Sol. $ax^2 + bx + c = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

Give $\alpha < \beta$, $0 < a < c$, $b = a + c$

$ax^2 + ax + cx + c = 0$

$ax (x + 1) + c (x + 1) = 0$

$(x + 1) (ax + c) = 0$

$x = -1, -\frac{c}{a}$

$\because$ $a > 0, c > 0 \Rightarrow b > 0$

$\because c > a \Rightarrow \frac{c}{a} > 1$

$-\frac{c}{a} < -1$

$\therefore$ B, D are correct.

MATCH THE COLUMN TYPE QUESTION(S)

46. Consider the quadratic trinomial $f(x) = 2x^2 - 10px + 7p - 1$, where p is a parameter. Find the range of p in the following conditions given in column-I :

Column-I	Column-II
(A) If both roots of f(x) = 0 are confined in (– 1, 1) then	(P) $\left(\frac{2}{5}, \infty\right)$
(B) Exactly one root of f(x) = 0 lies in (–1, 1)	(Q) ϕ
(C) Both roots of f(x) = 0 are greater than 1	(R) $\left(-\frac{1}{17}, \frac{1}{3}\right)$
(D) One root of f(x) = 0 is greater than 1 and other root of f(x) = 0 is less than – 1	(S) $\left(-\infty, -\frac{1}{17}\right] \cup \left[\frac{1}{3}, \infty\right)$

(T) $\left(-\infty,-\frac{1}{17}\right)\cup\left(\frac{1}{3},\infty\right)$

द्विघात त्रिपद $f(x) = 2x^2 - 10px + 7p - 1$, जहाँ p एक प्राचल है। दी गई शर्तों के अनुसार p के परिसर का मिलान कीजिए :

Column-I	**Column-II**
(A) यदि $f(x) = 0$ के दोनों मूल अन्तराल $(-1, 1)$ में स्थित हो तो	(P) $\left(\frac{2}{5}, \infty\right)$
(B) $f(x) = 0$ का एक मूल अन्तराल $(-1, 1)$ में स्थित हो	(Q) ϕ
(C) $f(x) = 0$ के दोनों मूल एक से अधिक हो	(R) $\left(-\frac{1}{17},\frac{1}{3}\right)$
(D) $f(x) = 0$ का एक मूल एक से अधिक तथा अन्य मूल -1 से छोटा हो तो	(S) $\left(-\infty,-\frac{1}{17}\right]\cup\left[\frac{1}{3},\infty\right)$
	(T) $\left(-\infty,-\frac{1}{17}\right)\cup\left(\frac{1}{3},\infty\right)$

Ans. (A) R (B) S (C) Q (D) Q

Sol. $f(x) = 2x^2 - 10px + 7p - 1$

(A) If both roots of $f(x) = 0$ confined in $(-1,1)$

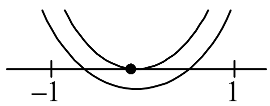
(i) $D \ge 0$

$100p^2 - 8(7p - 1) \ge 0$

$4[25p^2 - 14p + 2] \ge 0$

$25p^2 - 14p + 2 \ge 0$

$\because$ $D < 0, P \in R$



(ii) $-1 < \frac{-b}{2a} < 1$

$-1 < \frac{10p}{4} < 1$

$-\frac{2}{5} < p < \frac{2}{5}$

$p \in \left(-\frac{2}{5},\frac{2}{5}\right)$

(iii) $a \cdot f(1) > 0$

$2\cdot\{2 + 10p + 7p - 1\} > 0$

$17p + 1 > 0$

$p > -\frac{1}{17}$

(iv) $a \cdot f(1) > 0$

$2\cdot\{2 - 10p + 7p - 1\} > 0$

$-3p + 1 > 0$

$3p - 1 < 0$

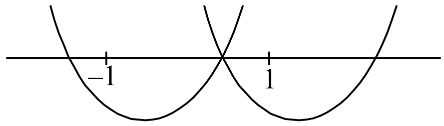
$p < \frac{1}{3}$

Now (i) $\cap$ (ii) $\cap$ (iii) $\cap$ (iv)

$p\left(-\frac{1}{17}, \frac{1}{3}\right)$

$A \to R$

(B) Exactly one root of $f(x) = 0$ lies in $(-1, 1)$



Case–I :

$D > 0$

$p \in R$

$f(-1) < 0$

$2 + 10p + 7p - 1 < 0$

$17p + 1 < 0$

$p < -\frac{1}{17} \Rightarrow p \in \left(-\frac{1}{17}, \infty\right)$

Case–II :

$D > 0 \Rightarrow p \in R$

$p(1) < 0$

$2 - 10p + 7p - 1 < 0$

$1 - 3p < 0$

$p > \frac{1}{3} \Rightarrow p \in \left(\frac{1}{3}, \infty\right)$

$(i) \cup (ii)$

$p \in \left(-\infty, -\frac{1}{17}\right) \cup \left(\frac{1}{3}, \infty\right)$

for $p = -\frac{1}{17} \Rightarrow$

$$f(x) = 2x^2 - 10x\left(-\frac{1}{17}\right) + 7\left(-\frac{1}{17}\right) - 1$$

$$= 2x^2 + \frac{10}{17}x - \frac{7}{17} - 1$$

$$= 2x^2 + \frac{10}{17}x - \frac{24}{17}$$

$$= \frac{1}{17}\left[34x^2 + 10x - 24\right]$$

$$= \frac{2}{17}\left[17x^2 + 5x - 12\right]$$

$$=\frac{2}{17}\left[17x^2+17x-12x-12\right]$$

$$=\frac{2}{17}\left[17x(x+1)-12(x+1)\right]$$

$$=\frac{2}{17}(x+1)(17x-12)$$

for $p=\frac{1}{3}$

$$f(x)=2x^2-10x\cdot\left(\frac{1}{3}\right)+7\left(\frac{1}{3}\right)-1$$

$$=2x^2-\frac{10x}{3}+\frac{4}{3}$$

$$=\frac{1}{3}\left[6x^2-10x+4\right]=\frac{2}{3}\left[3x^2-5x+2\right]$$

$$=\frac{2}{3}\left[3x^2-3x-2x+2\right]=\frac{2}{3}\left[3x(x-1)-2(x-1)\right]$$

$$=\frac{2}{3}(x-1)(3x-2)$$

$\because p=-\frac{1}{17}$ & $\frac{1}{3}$ satisfying the condition

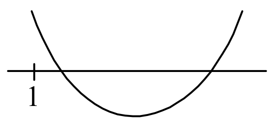
$\therefore p=-\frac{1}{17},\frac{1}{3}$ includes $p\in\left(-\infty,-\frac{1}{17}\right]\cup\left[\frac{1}{3},\infty\right)$

$B\rightarrow S$

(C) Both roots of f(x) = 0 are greater than 1

(i) $D>0\Rightarrow p\in R$

(ii) $f(1)>0$



$2-10p+7p-1>0$

$1-3p>0$

$$p<\frac{1}{3}\Rightarrow p\in\left(-\infty,\frac{1}{3}\right)$$

(iii) $-\frac{b}{2a}>1$

$$\frac{10p}{4}>1$$

$$p>\frac{2}{5}$$

$$p\in\left(\frac{2}{5},\infty\right)$$

$p \in \phi$

$C \to Q$

(D) One root of f(x) = 0 is greater than 1 and other root of f(x) = 0 is less than –1.

(i) $f(-1) < 0$

$2 + 10p + 7p - 1 < 0$

$17p + 1 < 0$

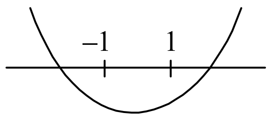
$p < -\frac{1}{17}$

(ii) $f(1) < 0$

$2 - 10p + 7p - 1 < 0$

$1 - 3p < 0$

$p > \frac{1}{3}$



(i) $\cap$ (ii)

$p \in \phi$

$D \to Q$

47. The roots of the quadratic equation, $x^2 - 5x + 6 = 0$ is α and β. Then find the following :

Column-I	**Column-II**
(A) $\alpha - \beta$	(P) 1
(B) $\alpha^2 + \beta^2$	(Q) – 13
(C) $\frac{\alpha^3 + \beta^3}{35}$	(R) – 1
(D) $\frac{\alpha^4 + \beta^4 - 62}{\alpha^3 + \beta^3}$	(S) 13

समीकरण $x^2 - 5x + 6 = 0$ के मूल α तथा β है। तब निम्न का मान प्राप्त करे :

Column-I	**Column-II**
(A) $\alpha - \beta$	(P) 1
(B) $\alpha^2 + \beta^2$	(Q) – 13
(C) $\frac{\alpha^3 + \beta^3}{35}$	(R) – 1
(D) $\frac{\alpha^4 + \beta^4 - 62}{\alpha^3 + \beta^3}$	(S) 13

Ans. (A) PR (B) S (C) P (D) P

Sol. $x^2 - 5x + 6 = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$

$\alpha + \beta = 5$

$\alpha\beta = 6$

(A) $|\alpha-\beta| = \left|\frac{\sqrt{D}}{a}\right| = \left|\frac{\sqrt{25-24}}{1}\right| = 1 \Rightarrow (\alpha-\beta) = \pm 1$

$A \to P, R$

(B) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 25 - 12 = 13$

$B \to S$

(C) $\frac{\alpha^3+\beta^3}{35} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{35} = \frac{125 - 3\times 6\times 5}{35} = \frac{125-90}{35}$

$= \frac{35}{35} = 1$

$C \to P$

(D) $\frac{\alpha^4+\beta^4-62}{\alpha^3+\beta^3} = \frac{(\alpha^2+\beta^2)^2 - 2\alpha^2\beta^2 - 62}{35} = \frac{(13)^2 - 2\times 36 - 62}{35}$

$= \frac{169-72-62}{35} = \frac{169-134}{35} = \frac{35}{35} = 1$

$D \to P$

48. Let $f(x) = \frac{x^2-6x+5}{x^2-5x+6}$: **[JEE-2007]**

Column-I	**Column-II**
(A) If $-1 < x < 1$, then f(x) satisfies	(P) $0 < f(x) < 1$
(B) If $1 < x < 2$, the f(x) satisfies	(Q) $f(x) < 0$
(C) If $3 < x < 5$, then f(x) satisfies	(R) $f(x) > 0$
(D) If $x > 5$, then f(x) satisfies	(S) $f(x) < 1$

माना $f(x) = \frac{x^2-6x+5}{x^2-5x+6}$: **[JEE-2007]**

Column-I	**Column-II**
(A) If $-1 < x < 1$, then f(x) satisfies	(P) $0 < f(x) < 1$
(B) If $1 < x < 2$, the f(x) satisfies	(Q) $f(x) < 0$
(C) If $3 < x < 5$, then f(x) satisfies	(R) $f(x) > 0$
(D) If $x > 5$, then f(x) satisfies	(S) $f(x) < 1$

Ans. (A) PRS (B) QS (C) QS (D) PRS

Sol. Let $f(x) = \frac{x^2-6x+5}{x^2-5x+6} = \frac{(x-5)(x-1)}{(x-2)(x-3)}$

Let $y = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

(i) $f(x) < 0$

$$\frac{(x-5)(x-1)}{(x-2)(x-3)} < 0$$

$x \in (1,2) \cup (3,5)$

(ii) $f(x) < 1$

$$\frac{x^2 - 6x + 5}{x^2 - 5x + 6} - 1 < 0$$

$$\frac{x^2 - 6x + 5 - x^2 + 5x - 6}{(x-2)(x-3)} < 0$$

$$-\frac{(x+1)}{(x-2)(x-3)} < 0$$

$$\frac{(x+1)}{(x-2)(x-3)} > 0$$

$x \in (-1,2) \cup (3,+\infty)$

(iii) $f(x) > 0$

$x \in (-\infty,1) \cup (2, 3) \cup (5, \infty)$

(iv) $0 < f(x) < 1$

(ii) $\cap$ (iii)

$x \in (-1, 1) \cup (5, \infty)$